偏导数: 设 $P_0(x_0,y_0)\in R^2$, 函数 $z=f(x,y)$ 在 P_0 的 δ 邻域 $N_\delta(P_0)$ 内有定义, 在 $N_\delta(P_0)$ 中固定 $y=y_0$
得到一元函数 $f(x,y_0)$, 若 $f(x,y_0)$ 在 x_0 处可导, 即

$$\lim_{\Delta x\to 0}\frac{f(x_0+\Delta x,y_0)-f(x_0,y_0)}{\Delta x}$$

写法 $f_x'(x_0,y_0)$ $\quad f_1'(x_0,y_0)$ $\quad \frac{\partial f}{\partial x}(x_0,y_0)$ $\quad \left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)}$

$f(x,y,z)=x^y+y^z+z^x \quad x,y,z>0$

求 $\frac{\partial f}{\partial x}$

$y\cdot x^{y-1}+e^{x\ln z}\cdot\ln z$

设函数 $z=f(x,y)$ 在 $P_0(x_0,y_0)$ 的某邻域 $N_\delta(P_0)$ 内偏导, 且 $f_x'(x,y)$. $f_y'(x,y)$ 在 $N_\delta(P_0)$ 内有界
则 $f(x,y)$ 在 P_0 处连续

$$f(x,y)-f(x_0,y_0)=|f(x,y)-f(x,y_0)|+|f(x,y_0)-f(x_0,y_0)|$$

$$=\frac{\partial f}{\partial y}(x,\eta)(y-y_0)$$

$$\frac{\partial f}{\partial x}(\xi,y_0)(x-x_0)$$

$(x,y)\to(x_0,y_0)$

设 $f(x,y)=x^2y+\sin(xy)+2e^x+y$，求 $f'_x(0,0)$，$f'_y(0,0)$，求 $f'_x(0,0)$，$f'_y(0,0)$

① 定义：$f'_x(0,0)=\lim\limits_{x\to 0}\dfrac{f(x,0)-f(0,0)}{x}=\dfrac{2e^x-2}{x}=2$

$f'_y(0,0)=\lim\limits_{y\to 0}\dfrac{f(0,y)-f(0,0)}{y}=\dfrac{2+y-2}{y}=1$

② 偏导公式 在该点有定义、导数存在

$f'_x(x,y)=2xy+\cos(xy)\cdot y+2e^x \Rightarrow f'_x(0,0)=2$

$f'_y(x,y)=x^2+\cos(xy)\cdot x+1 \Rightarrow f'_y(0,0)=1$

$f(x,y)=\arctan\frac{x}{y}+\ln(2x+y)+y^4$ 求 $\frac{\partial f}{\partial x}$

$=\dfrac{\frac{1}{y}}{1+(\frac{x}{y})^2}+\dfrac{2}{2x+y}=\dfrac{y}{x^2+y^2}+\dfrac{2}{2x+y}$

二阶混合偏导

$f''_{xy}(x_0,y_0)=\lim\limits_{y\to y_0}\dfrac{f'_x(x_0,y)-f'_x(x_0,y_0)}{y-y_0}$

$f''_{yx}(x_0,y_0)=\lim\limits_{x\to x_0}\dfrac{f'_y(x,y_0)-f'_y(x_0,y_0)}{x-x_0}$

求 $f(x,y)=\begin{cases} xy\dfrac{x^2-y^2}{x^2+y^2} & (x,y)\neq(0,0) \\ 0 & (x,y)=(0,0) \end{cases}$

$$f''_{xy}(0,0)=\lim_{y\to 0}\frac{f_x(0,y)-f_x(0,0)}{y-0}$$

$$f_x(0,0)=\lim_{x\to 0}\frac{f(x,0)-f(0,0)}{x-0}=0$$

$$f_x(0,y)=\lim_{x\to 0}\frac{f(x,y)-f(0,y)}{x-0}$$

$$=\lim_{x\to 0} y\frac{x^2-y^2}{x^2+y^2}=-y$$

若二阶混合偏导 $f''_{xy}(x,y)$ 与 $f''_{yx}(x,y)$ 处皆连续，则 $f''_{xy}(x,y)=f''_{yx}(x,y)$.

$v:\quad y\mapsto f(x,y)-f(0,y)$

$u:\quad x\mapsto f(x,y)-f(x,0)$

$\Delta:=\{f(x,y)-f(x,0)\}-\{f(0,y)-f(0,0)\}$

$$\Delta=u(x)-u(0)=u'(\xi)x$$
$$=x\{f_x(\xi,y)-f_x(\xi,0)\}$$
$$=xy\cdot f_{xy}(\xi,\eta)$$

按 $y.\{f_y(x,\eta)-f_y(0,\eta)\}\cdot y.$ 无法继续

$$\Delta=\{f(x,y)-f(0,y)\}-\{f(x,0)-f(0,0)\}$$
$$=v(y)-v(0)=v'(t)y$$
$$=y\{f_y(x,t)-f(0,t)\}$$
$$=yx\cdot f_{yx}(s,t)$$

当 $(x,y)\to(0,0)$ 时 $(\xi,\eta),(s,t)\to(0,0)$